Project Plan

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MGMT6219 Data Analytics Capstone

[redacted]

Due Date: March 7th, 2026

Working in your assigned team, develop a Project Plan (approximately 5-7 pages including appendices) that clearly explains how you will approach your Riipen final project. Your plan must include:

Problem Statement & Research Questions (1-2 pages):

- One concise business challenge summary written in your own words.
  - [redacted] is in need of a dashboard template for administrators at their consulting agency. They are lacking a product that can offer high-level strategic insights for strategic purposes to scale and grow the business more effectively.

- 5-7 specific, measurable, analytical questions you will answer.
  - Can our machine learning model predict next month's revenue with less than 10% error to help with financial planning?
  - Can the dashboard show the exact percentage of income from our top client to help the manager find 2-3 new customers?
  - If we lose our biggest customer today, what percentage of our total monthly revenue would disappear, and how fast can we replace it?
  - Which specific consulting services have the highest profit margins based on our time logs?
  - What kind of level of security should we use in analytics tools?
  - Does our new pricing model give the same recommendations for similar clients without any unfair bias?

- A brief note on how the project connects to your concurrent courses (e.g., Predictive Analytics, Data Strategy, Ethics)
  - This project connects to our concurrent courses by taking the elements learned from each class—Predictive Analytics, Data Strategy, Ethics, AI and Machine Learning, and Data Analytics for CRM—and implementing them to create a finished deliverable that is ready for market.
    - Predictive Analytics
      - Predictive forecasting and modelling
    - Data Strategy
      - Strategies for governance, framing the business question, establishing data goals and objectives, identifying KPIs, establishing data infrastructure and architecture, carefully selecting data and how it will be used, identifying a data-driven decision-making framework, highlighting the roadmap for implementation, and identifying risks and mitigations
    - Ethics
      - Carefully following regulations, identifying and mitigating biases and other ethical concerns, and establishing governance practices
      - Adhere to the confidentiality agreement that we each signed
    - AI and Machine Learning
      - Potentialty utilizing various techniques for text mining, sentiment analysis, and clustering

        - Data Analytics for CRM
            - Implementing data mining techniques

Methodology & Methods Selection (1-2 pages):

- A clear explanation of the analytical approaches you will use (e.g., descriptive, predictive, statistical testing, segmentation) and why they are appropriate.
    - Descriptive analysis: to organize the dataset
    - Predictive analysis: build models to predict next month's revenue and when customer will pay
    - Prescriptive analysis: provide recommendations on pricing and business growth
    - Segmentation: we have not received the dataset yet, so we are not sure if we need to use segmentation for clustering customers.

- A description of your planned data sources (Riipen data plus any supplemental open data, if applicable) and how you will access them.
    - Upon receiving the data, we immediately verify its completeness and accuracy to ensure client privacy. If necessary, we will consult publicly available websites for average prices in the Canadian consulting industry.

- Identification of tools and technologies (e.g., Python, SQL, Power BI, ML tools) and how each will be used in the project.
    - Visualization: using Power BI to connect the cleaned dataset from python
    - Time Series: to predict the revenue forecasting and payment timing prediction
    - Python: Used for "data wrangling" and building the automated pipeline. Python is best for cleaning "messy" data and ensuring accuracy and completeness.

- A high-level timeline highlighting major phases (data prep, analysis, visualization, reporting, and presentation).
    - Week 9 Project Plan: meeting Ray, kick off planning
    - Week 10 Data Prep & Ethics due Mar 11: Complete data dictionary (tabular format); Assess data quality report; clean data
    - Week 11 De 2. Predictive Analytics Models: Revenue forecasting; Payment timing prediction, and Client segmentation for strategic planning
    - Team Progress check-in due March 18
    - Week 12 De 3. Interactive Power BI: Cash flow dashboard; real-time business metrics
    - Week 13 final report & dashboard due Apr. 1
    - De4. Strategic Analysis Report: Recommends pricing optimization & growth; Deployed free-tier cloud platforms; Tested real business operations
    - Week 14 final pre-in-class Apr. 8: Require the feedback
    - Week 15: Submit GitHub Apr. 15

Project Plan Artifacts (2-3 pages):

- Gantt chart showing tasks, owners, and key deadlines from Weeks 8–15.

- Team roles and responsibility matrix (who is lead for data, methods, visualization, communication, QA, etc.).

Team Member	Role	Responsibility
Jason	Coding Lead Communication Lead	1. Works in tandem with data lead to ensure data is effectively analyzed using organized, and efficient code 2. Transforming data into actionable insights and recommendations 3. Develops a communication plan that maps out stakeholders and meetings, determines appropriate channels, sets communication cadence, etc. 4. Acts as a bridge between the Riipen project lead and the data team to communicate business needs and technical aspects
[redacted]	Visualization Lead	1. To identify the KPIs at 8-9. 2. To communicate data formats with the data lead, then plan the script of dashboard (need CSV or excel file; what's the format for output from predictive models?) 3. To ensure the real-time & interactive in dashboard, like can reflect the python pipeline for scraping the updated stripe payment status 4. To make the dashboard readable
[redacted]	Analytics lead & insights strategist	1. Lead the development of analytical models, ensuring appropriate methods. 2. Translate data into actionable insights connecting findings to project goals. 3. Collaborate with the data lead and visualization lead, becoming a bridge between data and outputs.
[redacted]	Data Lead & Wrangler Quality Assurance Lead	1. Ensure data is clean and well organized 2. Standardize formats 3. Communicate categorization and formatting with a data dictionary
[redacted]	Governance & Implementation Lead	1. Ensure the project follows MGMT6215 rules 2. Monitor the risk checklist and track the mitigation steps weekly (data delays, tools issues) 3. Test the deployment, like help linya to test if the dashboard works correctly on different devices before the final demo

- Risk assessment and mitigation strategies (e.g., data access delays, tool issues, team availability).

We have classified the risk in 2 major areas for this project: operational and strategic.

Operational Risk:

1. **Data Quality Risk**

The system relies on time logs to generate invoices and forecasts. If the time logs contain missing values, incorrect timestamps, or inconsistent formats, the automated billing pipeline may generate incorrect payment links or inaccurate forecasts and can impact incorrect invoices sent to the clients and inaccurate forecasts for the revenue.

Mitigation:

- To reduce the risk in data quality, the team must ensure a cleaning process with the following rules:
- Implement data validation rules
- Require mandatory fields such as client name, date, billing name, payment term, and others needed to develop the project.
- Implement a flagged review step for manual entries.

2. **API integration Risk**

The project depends on Stripe API and email automation. API changes, authentication failures, or connection issues could prevent payment links from being generated or sent, creating delays, failed transactions, or an interrupted workflow automation.

Mitigation:

- Implement API error handling when doing the Python code.
- Monitor API responses and failures, as well as the likelihood and frequency.
- Maintain the manual billing process in case of a failure with the API.

3. **Data Security and Privacy Risk**

The system contains all types of sensitive information, such as client names, billing information, financial transactions, etc., that may be exposed if not secured properly.

Mitigation:

- Restrict access to data only for people involved in the billing process.
- Possible use of encrypted communication for all API transactions.
- Use secure API keys

4. **Dashboard Latency or data refresh failures (Timeliness)**

Software like Power BI, when it comes to dashboarding, may not update in real time due to refresh failures or cloud resource limits and can impact any decision-making process.

Mitigation:

- Schedule refreshes with buffer time.
- Keep visualizations lightweight and optimized.
- Use incremental refreshes where possible.

5. **Dashboard Interpretation Risk**

Selecting the correct charts to plot any graph and the user that will be reading the dashboard is important; if the dashboard is not clear, users may misinterpret dashboard insights or predictive analytics outputs when making business decisions.

Mitigation:

- Include clear labels and explanations in the dashboard.
- Provide summary insights rather than the raw metrics alone.
- Select proper graphs to show that facilitate the decision-making process.
- Include documentation describing how predictions should be interpreted.

Data Governance & Ethics Checklist

- Data governance and ethics checklist indicating how you will protect confidentiality, handle sensitive data, and align with MGMT-6215 principles.
    - Yet to be given a dataset or details about it.
    - Data Classification
        - Personal identify information, like email address, name and sensitivity personal information
        - Crown Jewels: these are core business operation data, such as payment records, Stripe transaction data, and consulting business metrics, and these are needed for security at a high level.
    - Clearly identify purpose
    - Meaningful Consent
    - Transparency
        - Ensuring consultants can understand the decisions that are made by the models.
    - Safeguards & Monitoring
        - C (confidentiality); I (Integrity); A (Availability)
    - Bias & Fairness
        - Evaluate if the model's output is fair to all customers.
    - Data Quality

- During week 10, our team are going to focus on data cleaning, so we need to check data accuracy, completeness, and consistency to avoid errors caused by dirty data